

# NELSON DANG

778-955-8538 | [NELSONDANG14@GMAIL.COM](mailto:NELSONDANG14@GMAIL.COM) | [LINKEDIN.COM/IN/NELSON-DANG-6409B5386](https://www.linkedin.com/in/nelson-dang-6409b5386) | [GITHUB.COM/NELSONDG](https://github.com/NELSONDG) | [NELSONDANG.DEV](https://NELSONDANG.DEV)

## EDUCATION

### Douglas College

*Diploma in Computer Science*

New Westminster, BC

*Sep. 2023 – Sep. 2025*

### Simon Fraser University

*Bachelor's in Computer Science*

Surrey, BC

*Jan. 2026 – Expected April 2029*

## PROJECTS

### Tochi Storage App | *Next.js, TypeScript, React, Supabase, PostgreSQL, Tailwind CSS*

May 2026 – Present

- Built a full-stack inventory and finance management web app for a food truck business, replacing manual spreadsheet tracking and reducing bookkeeping time by about 70%
- Designed a real-time finance dashboard tracking revenue, cost of goods, gross profit, and margins with automated COGS calculation from inventory stock usage
- Implemented role-based access control with three permission levels (admin, manager, staff) using Supabase Auth and PostgreSQL row-level security policies
- Integrated live inventory tracking with low-stock alerts, stock adjustment history, and automatic cost calculations tied to item usage

### Escape Prison Break Game | *Java, Maven, JUnit*

Jan 2026 – April 2026

- Built a 4-level 2D grid-based prison escape game in Java with a Swing GUI, featuring pixel-art sprite rendering, animated walk cycles, and a player-centered scrolling camera viewport
- Implemented a breadth-first search pathfinding algorithm driving enemy AI state transitions between patrol and pursuit, triggered by a 5-tile line-of-sight check across a 4-level grid
- Applied the Factory Method design pattern to decouple all level configuration (spawn coordinates, coin requirements, exit positions) from the game engine, enabling clean addition of new levels
- Wrote 22 JUnit test suites achieving about 80% coverage across the game engine, board, score manager, entity classes, and all four level factories

### Mental Health Chatbot | *Python, Flask, SQLite, Google Gemini API, JavaScript*

Jun 2026 – Jul 2026

- Built a full-stack mental health support chatbot that classifies user messages into 20+ emotional states using VADER sentiment analysis, keyword scoring, and spaCy noun-phrase extraction
- Engineered an LLM integration layer over the Gemini REST API, injecting rolling conversation history and classified emotional state into structured prompts, with a deterministic guardrail routing crisis-flagged messages outside the model entirely
- Designed a crisis-detection safety net using word-boundary phrase matching with an idiom-exception list to reduce false positives, routing flagged messages to a hardcoded crisis-hotline response independent of the AI model
- Implemented voice input/output with the Web Speech API, a fully responsive dark-mode-aware UI, and a Flask/SQLAlchemy backend persisting conversation history per session

## EXPERIENCE

### Manager

May 2024 – Present

*Tochi Desserts LTD.*

*Vancouver, BC*

- Cut weekly bookkeeping 70% (5 hrs → 1.5 hrs) for a 3-person team by designing and deploying a full-stack Next.js/PostgreSQL inventory platform that replaced manual spreadsheet tracking
- Directed staffing, supply, and transport logistics for 6–10 events monthly — including multi-day festivals (FIFA Fan Festival, PNE, Christmas markets) — sustaining 100+ consecutive events with zero ingredient stockouts
- Owned end-to-end truck operations in the owner's absence, leading 8 staff through peak service and reconciling about 450+ point-of-sale transactions per event

## TECHNICAL SKILLS

**Languages:** Python, JavaScript, TypeScript, Java, C++, SQL, HTML/CSS

**Frameworks:** React, Flask, Next.js

**Developer Tools:** Node.js, Git, VS Code, Visual Studio, JetBrains, Vercel, Supabase, JUnit, Maven

**Interests:** Weight lifting, Video games, Running, Boba